

MANIPAL UNIVERSITY JAIPUR
Department Artificial Intelligence and Machine Learning Course
Hand-out

A. Basic Details:

Programme Name:	BTech CSE(AI ML)
Course Name:	Recommender System
Course Code:	AIM3143
LTPC (<i>Lecture Tutorial Practical Credits</i>):	3 0 0 3
Session:	Jul-Nov 2026
Class:	V
Course Coordinator:	Dr. Sonia
Course Instructor(s):	Dr. Surendra Solanki / Dr. Pranshu Pranjali / Dr. Sonia Choudhary / Ms. Akanksha Mrinali / Ms. Rishika / Dr. Neeetu Gupta / Dr. Neha Mahala / Dr. Shivendra Dubey / Dr. Pramod Kumar

B. Introduction: This course provides a comprehensive introduction to Recommender Systems, essential in machine learning and data analytics for predicting user preferences from large datasets. Widely used in e-commerce, social media, and content streaming, these systems enhance user experience and engagement. The course covers fundamental concepts and techniques in designing and evaluating recommender systems, exploring collaborative filtering, content-based, knowledge-based, and hybrid approaches.

- a) Core functions include understanding linear algebra foundations like matrix operations.
- b) Apply user-based and item-based collaborative filtering and learn about their vulnerabilities.
- c) Explore content-based systems' architecture, feature extraction, and classification algorithms.
- d) Gain insights into knowledge-based systems, examining representation and reasoning techniques.
- e) Investigate hybrid approaches, understanding design strategies and limitations.

C. Course Outcomes :

CO Statement	CO	Bloom's Taxonomy Level	Target Attainment (%)	Target Attainment Level
Explain the fundamentals of recommender systems and recommendation techniques.	AIM3143.1	Cognitive: Understand	90%	2.7
Apply collaborative filtering and model-based recommendation techniques.	AIM3143.2	Cognitive: Apply	80%	2.4
Analyze preprocessing methods, security issues, and evaluation metrics in recommender systems.	AIM3143.3	Cognitive: Analyze	80%	2.4
Analyse content-based and knowledge-based recommender systems.	AIM3143.4	Cognitive: Analyze	80%	2.4
Evaluate hybrid recommender systems using appropriate recommendation and evaluation techniques.	AIM3143.5	Cognitive: Evaluate	80%	2.4

D. Program Outcomes and Program Specific Outcomes:

[PO.1] Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering tools to the solution of complex engineering problems.

[PO.2] Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

[PO.3] Design/development of solutions: Design solutions for complex engineering problems and

design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4] Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5] Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

[PO.6] The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

[PO.7] Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8] Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.G] Individual and Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10] Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11] Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12] Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES:

[PSO.1] Graduates will be able to examine the applications of Artificial Intelligence and Machine Learning in real-life problems.

[PSO.2] Graduates will be able to design and implement intelligent systems for multidisciplinary problems.

E. Assessment Plan

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Quiz (2): 10 Marks Case study/Assignments (2): 10 Marks Mini Project (max 2 student): 10 Marks	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

Attendance (Formative)	A minimum of 75% Attendance is required to be maintained by a student to be eligible to appear for the ETE. The allowance of 25% includes all types of leaves including medical leaves.
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F. Syllabus:

Introduction n: Recommender system functions, applications of recommendation systems, understanding ratings (explicit vs. implicit feedback), issues in recommender systems (sparsity, scalability, cold-start problem). **Mathematical Basic:** linear algebra concepts: Matrix representation, matrix multiplication, similarity measures (Cosine Similarity, Pearson Correlation). **Collaborative Filtering:** User-based nearest neighbor recommendation, Item-based nearest neighbor recommendation, limitations of memory-based methods. **Model-Based Approaches:** Matrix factorization techniques (SVD, ALS), latent factor models, handling sparsity and scalability, implicit feedback models. **Neural Collaborative Filtering:** Basic concepts and architecture. **Pre-processing Techniques:** Data normalization, handling missing values, dimensionality reduction. **Security and Robustness:** Attacks on collaborative recommender systems (Shilling attacks) and basic defense mechanisms. **Evaluation of collaborative filtering models.** **Content-Based recommendation:** High level architecture of content-based systems, Advantages and drawbacks of content-based filtering, Item profiles, discovering features of documents, obtaining item features from tags, representing item profiles, Methods for learning user profiles, Similarity based retrieval, Classification algorithms. **Knowledge based recommendation:** Knowledge representation and reasoning, Constraint based recommenders, Case based recommenders. **Hybrid approaches:** Opportunities for hybridization, **Monolithic hybridization design:** Feature combination, Feature augmentation, **Parallelized hybridization design:** Weighted, Switching, Mixed, Pipelined hybridization design: Cascade Meta-level, Limitations of hybridization strategies. **Evaluating Recommender System:** Introduction, General properties of evaluation research, Evaluation designs, Evaluation on historical datasets, Error metrics, Decision-Support metrics, User-Centred metrics.

References:

1. C.C. Aggarwal, Recommender Systems: The Textbook (1e), Springer, 2016.
2. N Manouselis, H. Drachsler, K. Verbert and E. Duval., Recommender Systems for Learning (1e), Springer 2013.
3. F. Ricci, L. Rokach, D. Shapira and B.P. Kantor, Recommender Systems Handbook (1e), Springer, 2011.

G. Lecture Plan:

Lecture No.	Topic	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing CO
1	Introduction to Recommender Systems	Understand recommender systems and their applications	AIM3143.1	Lecture based teaching-learning	Quiz
2	Functions and Applications of Recommendation Systems	Explain recommendation functions and applications	AIM3143.1	Lecture based teaching-learning	Assignment

3	Explicit vs. Implicit Feedback	Differentiate rating mechanisms	AIM3143.1	Lecture based teaching-learning	Quiz
4	Challenges: Sparsity, Cold-Start, Scalability	Analyze challenges in recommender systems	AIM3143.1	Lecture based teaching-learning	Assignment
5	Matrix Representation and Matrix Multiplication	Apply matrix representation in recommender systems	AIM3143.1	Lecture based teaching-learning	Numerical
6	Similarity Measures: Cosine Similarity & Pearson Correlation	Compute similarity measures	AIM3143.1	Lecture based teaching-learning	Problem Solving
7	User-Based Collaborative Filtering	Explain user-based collaborative filtering	AIM3143.2	Lecture based teaching-learning	Quiz
8	Item-Based Collaborative Filtering	Implement item-based recommendation	AIM3143.2	Lecture based teaching-learning	Assignment
9	Limitations of Memory-Based Methods	Analyze drawbacks of memory-based recommendation	AIM3143.2	Lecture based teaching-learning	Quiz
10	Model-Based Recommendation Systems	Understand model-based collaborative filtering	AIM3143.2	Lecture based teaching-learning	Assignment

11	Matrix Factorization using SVD	Apply Singular Value Decomposition	AIM3143.2	Lecture based teaching-learning	Numerical
12	Alternating Least Squares (ALS)	Explain ALS algorithm	AIM3143.2	Lecture based teaching-learning	Quiz
13	Latent Factor Models	Understand latent factor representation	AIM3143.2	Lecture based teaching-learning	Assignment
14	Handling Sparsity and Scalability	Explain techniques for sparse data	AIM3143.2	Lecture based teaching-learning	Problem Solving
15	Implicit Feedback Models	Analyze implicit feedback techniques	AIM3143.2	Lecture based teaching-learning	Quiz
16	Neural Collaborative Filtering	Understand Neural CF concepts and architecture	AIM3143.3	Lecture based teaching-learning	Assignment
17	Data Normalization	Apply normalization techniques	AIM3143.3	Lecture based teaching-learning	Practical
18	Handling Missing Values and Dimensionality Reduction	Preprocess recommendation datasets	AIM3143.3	Lecture based teaching-learning	Assignment
19	Security Issues in Recommendation Systems	Understand recommender system security	AIM3143.3	Lecture based teaching-learning	Quiz

20	Shilling Attacks and Defense Mechanisms	Analyze attacks and defenses	AIM3143.3	Lecture based teaching-learning	Case Study
21	Evaluation of Collaborative Filtering Models	Evaluate collaborative filtering models	AIM3143.3	Lecture based teaching-learning	Quiz
22	Introduction to Content-Based Recommendation	Understand content-based recommendation	AIM3143.4	Lecture based teaching-learning	Assignment
23	Architecture, Advantages and Drawbacks	Explain content-based architecture	AIM3143.4	Lecture based teaching-learning	Quiz
24	Item Profiles and Feature Discovery	Construct item profiles	AIM3143.4	Lecture based teaching-learning	Assignment
25	Tags, User Profiles and Similarity Retrieval	Apply similarity-based retrieval	AIM3143.4	Lecture based teaching-learning	Practical
26	Classification Algorithms for Content-Based Systems	Understand classification techniques	AIM3143.4	Lecture based teaching-learning	Quiz
27	Knowledge-Based Recommendation Systems	Understand knowledge-based recommendation	AIM3143.4	Lecture based teaching-learning	Assignment
28	Constraint-Based Recommendation	Apply constraint-based reasoning	AIM3143.4	Lecture based teaching-learning	Case Study

29	Case-Based Recommendation	Analyze case-based recommenders	AIM3143.4	Lecture based teaching-learning	Assignment
30	Hybrid Recommendation Systems	Understand opportunities for hybridization	AIM3143.5	Lecture based teaching-learning	Quiz
31	Monolithic Hybridization: Feature Combination & Feature Augmentation	Analyze monolithic hybrid methods	AIM3143.5	Lecture based teaching-learning	Assignment
32	Parallel Hybridization: Weighted, Switching & Mixed	Compare parallel hybrid approaches	AIM3143.5	Lecture based teaching-learning	Quiz
33	Pipelined Hybridization: Cascade & Meta-Level	Explain pipelined hybrid methods	AIM3143.5	Lecture based teaching-learning	Assignment
34	Limitations of Hybridization Strategies	Evaluate hybrid recommender limitations	AIM3143.5	Lecture based teaching-learning	Case Study
35	Evaluation Metrics: Error, Decision Support and User-Centered Metrics	Compute evaluation metrics	AIM3143.5	Lecture based teaching-learning	Numerical

36	Case Study and Course Review	Design and evaluate a recommender system	AIM3143.5	Lecture based teaching-learning	Presentation
End Term Examination					

H. Course Articulation Matrix:

CO	Statement	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
AIM3143.1	Explain the fundamentals of recommender systems and recommendation techniques.	3	3	1	1	2				2	1		2	3	2
AIM3143.2	Apply collaborative filtering and model-based recommendation techniques.	3	3	3	2	3				2	2		2	3	3
AIM3143.3	Analyze preprocessing methods, security issues, and evaluation metrics in recommender systems.	2	3	2	3	3	2		1	2	2		2	3	3
AIM3143.4	Analyze content-based and knowledge-based recommender systems.	2	3	2	3	3	2		1	2	2		2	3	3

AIM3143.5	Evaluate hybrid recommender systems using recommendation and evaluation techniques.															
		2	3	3	3	3			1	2	3	2	3	3	3	

1: Low Correlation 2: Moderate Correlation 3: Substantial Correlation

I. Learner-Centric Activities Low Difficulty Level Activities:

Peer-assisted learning on recommendation algorithms (Content-Based, Collaborative Filtering, Hybrid Filtering)

Recommended video tutorials for understanding recommendation workflows and evaluation metrics

Hands-on practice with similarity measures and basic recommendation datasets

Solving simple recommendation problems using Python libraries (Pandas, NumPy, Scikit-learn)

High Difficulty Level Activities

Case study-based assignments on recommendation systems used by Netflix, Amazon, YouTube, and Spotify

Mini projects on developing movie, music, book, or product recommendation systems

Hands-on implementation of user-based, item-based, matrix factorization, and hybrid recommender models

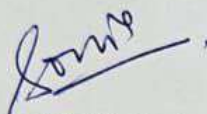
Performance evaluation using Precision, Recall, F1-score, MAP, and NDCG

MOOCs or certifications related to Recommender Systems, Personalization, and Recommendation Algorithms

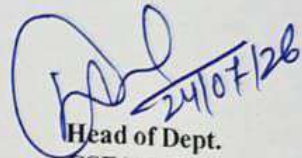
J. Outside Class Engagement Activities:

Activity	Difficulty Level	Hours Engaged
Participation in technical workshops, webinars, and seminars on Recommender Systems, Personalization, and AI applications	Medium	3 hrs
Group study and peer-learning sessions for collaborative filtering, content-based filtering, and evaluation metrics	Medium	2 hrs
Development of mini-projects and case studies on movie, music, e-commerce, or news recommendation systems	High	10 hrs

Completion of an online certification program – Recommender Systems and Personalization (Coursera, edX, Udemy, or similar platform)	High	15 hrs
Exploration and implementation of recommendation models using the MovieLens dataset or Amazon Product Review dataset	High	8 hrs
Participation in Kaggle competitions or practice notebooks on recommendation systems	High	6 hrs


Course Co-Ordinator


Student Representative
Name:
Registration No.:


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